

SHETH L.U.J & SIR M.V. COLLEGE

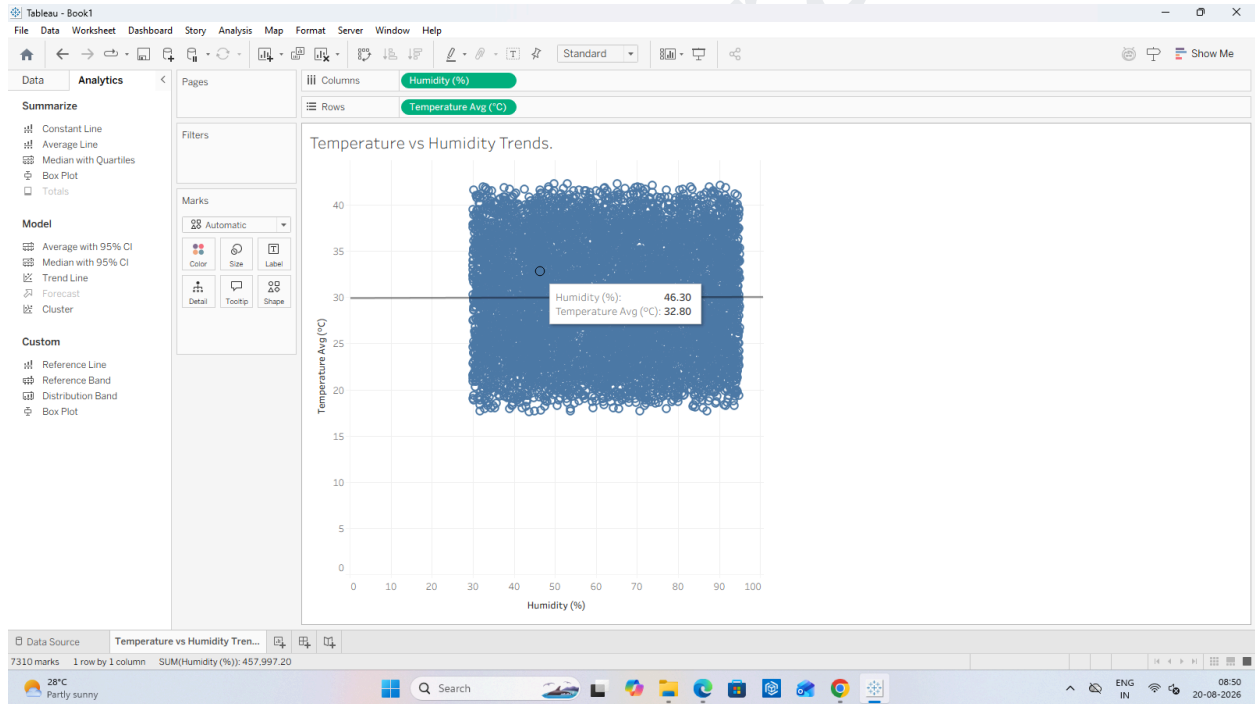
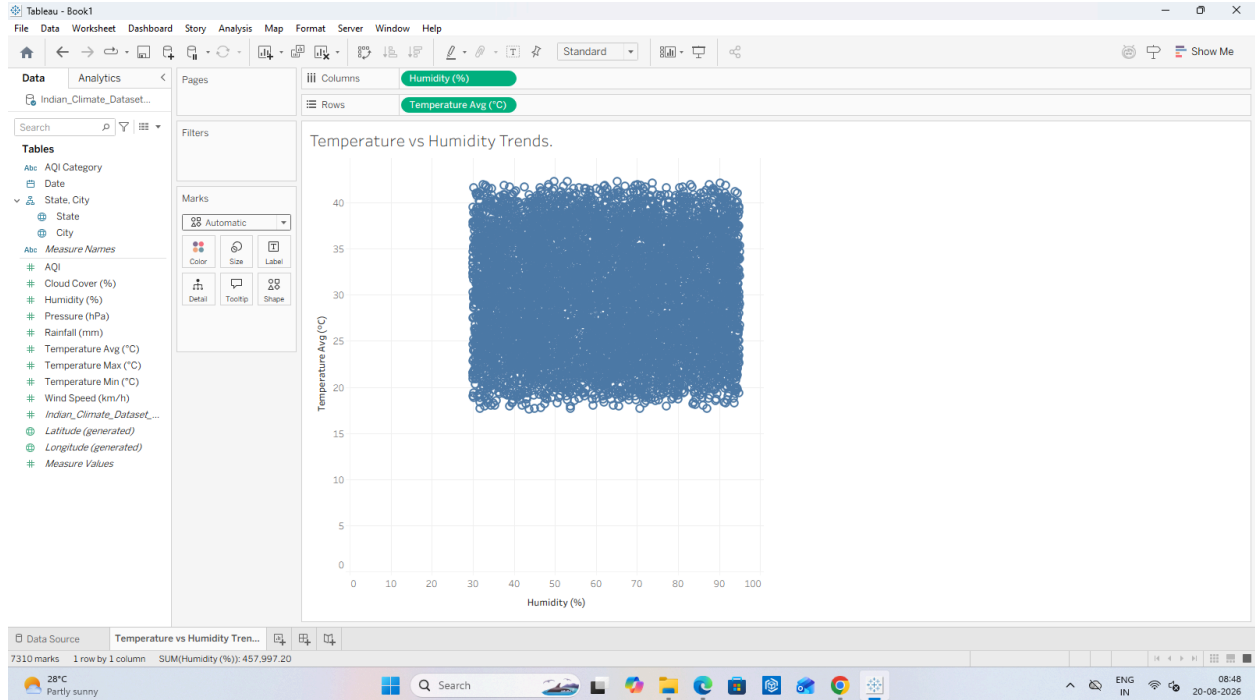
Aim: Performing Advanced Analytics using Trends, Clustering, and Forecasting

- a) Add and interpret trend lines (linear, exponential)
- b) Customize and analyze trend models
- c) Perform clustering on datasets
- d) Analyze distributions using built-in tools
- e) Apply forecasting techniques
- f) Compare different statistical models
- g) Export and interpret statistical results

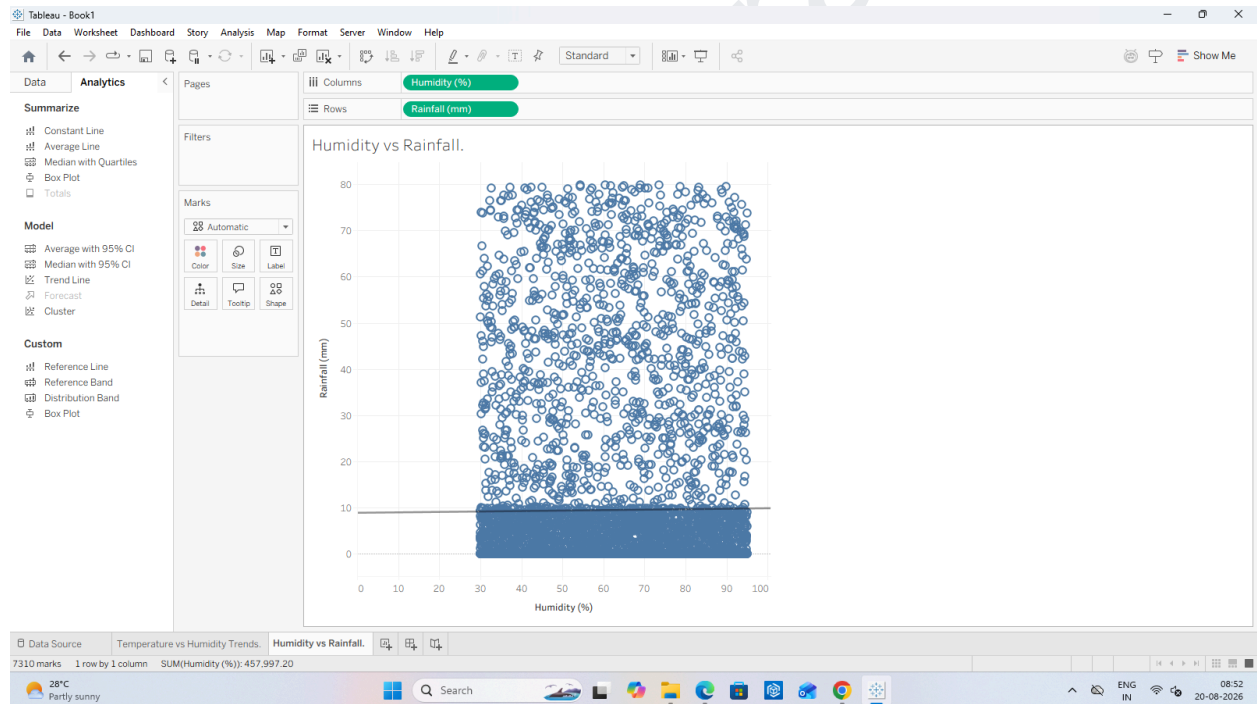
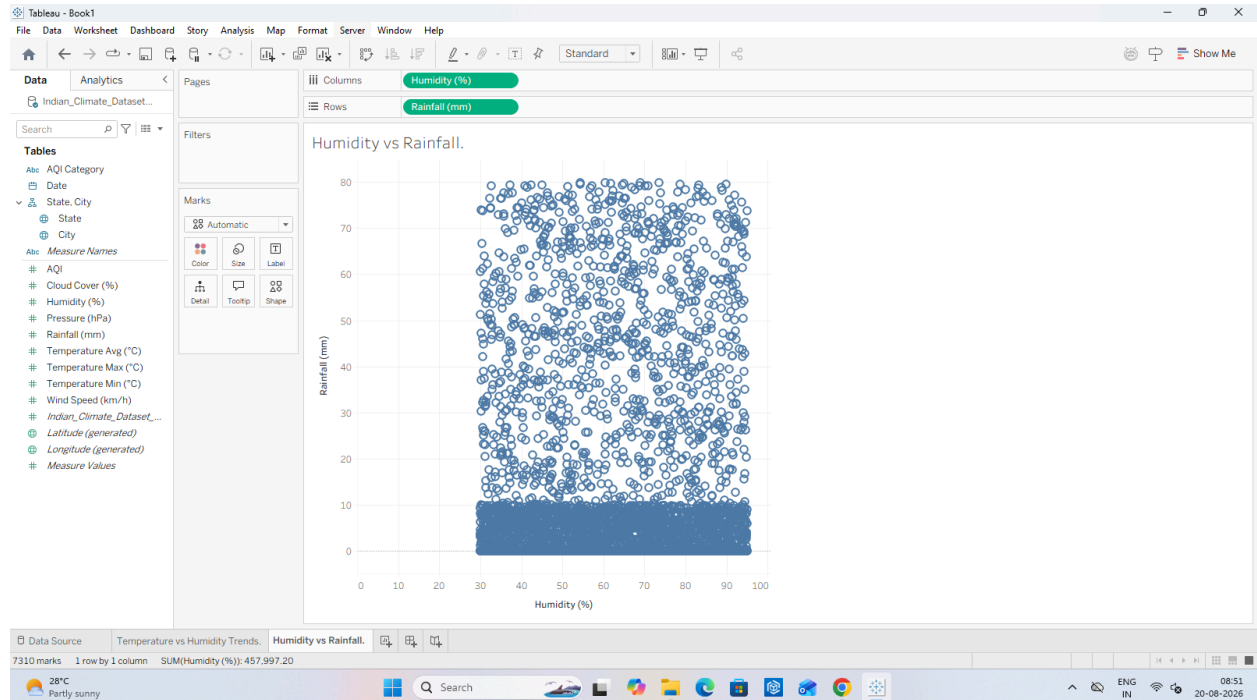
The screenshot displays the Tableau Desktop interface. The main view shows a table with the following data:

Date	City	State	Temperature Max (°C)	Temperature
01-01-2024	Mumbai	Maharashtra	32.5000	
01-01-2024	Delhi	Delhi	25.4000	
01-01-2024	Bengaluru	Karnataka	37.2000	
01-01-2024	Chennai	Tamil Nadu	37.2000	
01-01-2024	Kolkata	West Bengal	27.4000	
01-01-2024	Hyderabad	Telangana	44.4000	
01-01-2024	Ahmedabad	Gujarat	32.8000	

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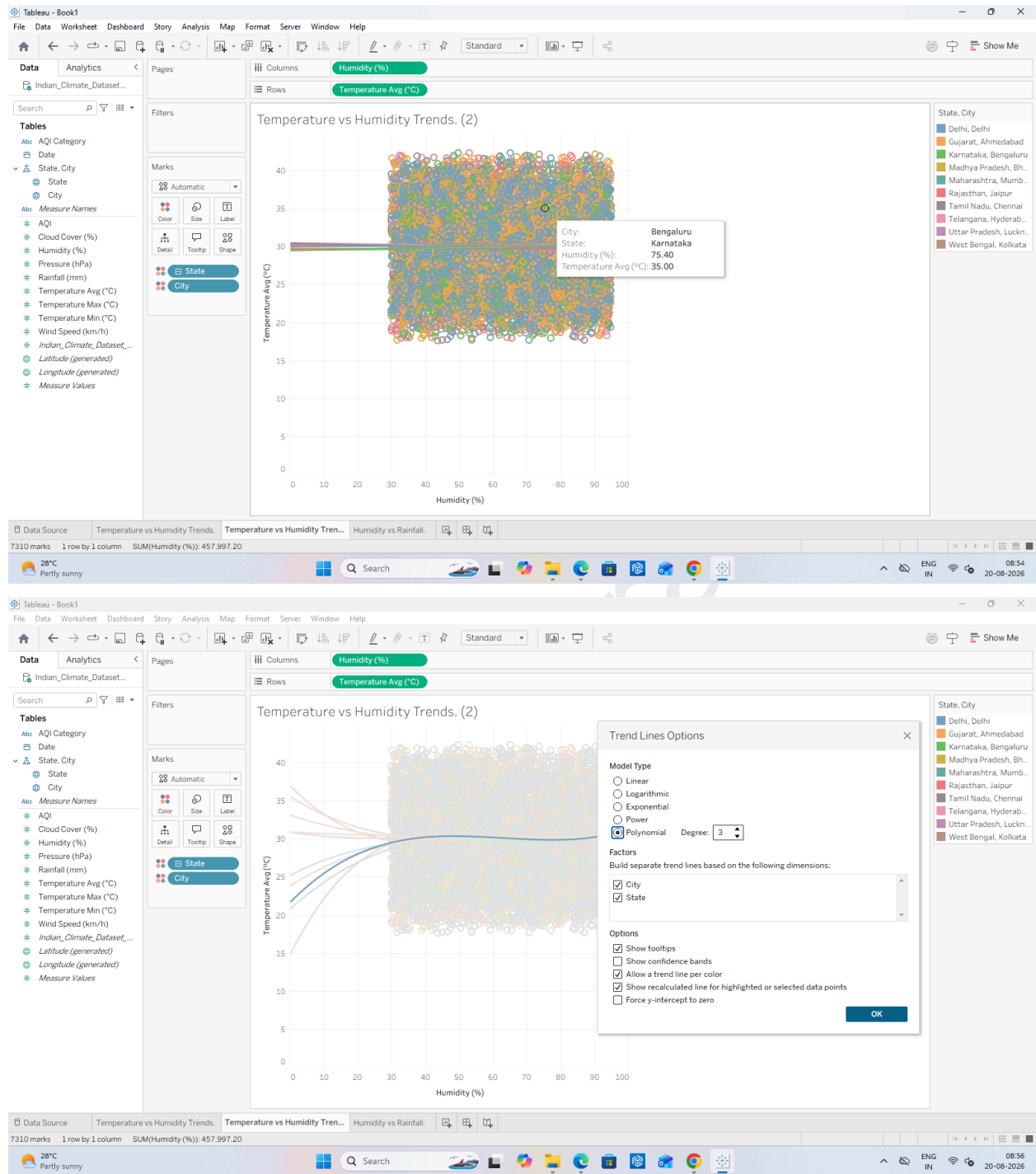
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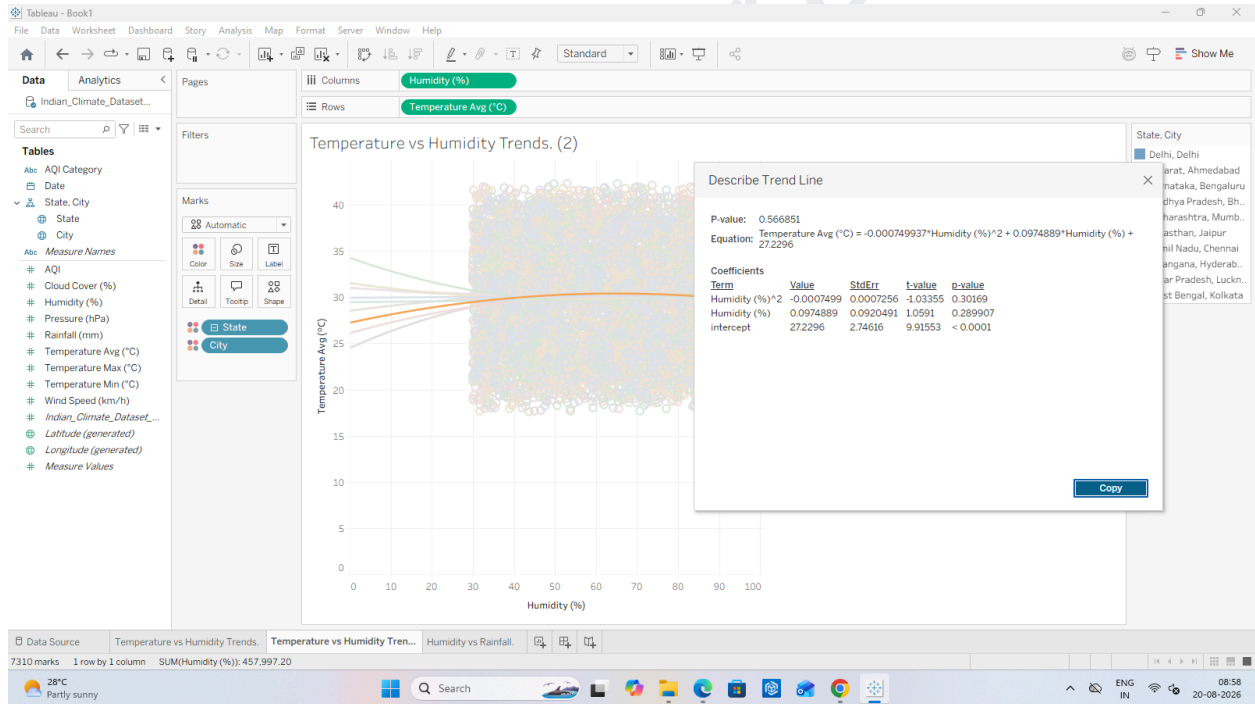
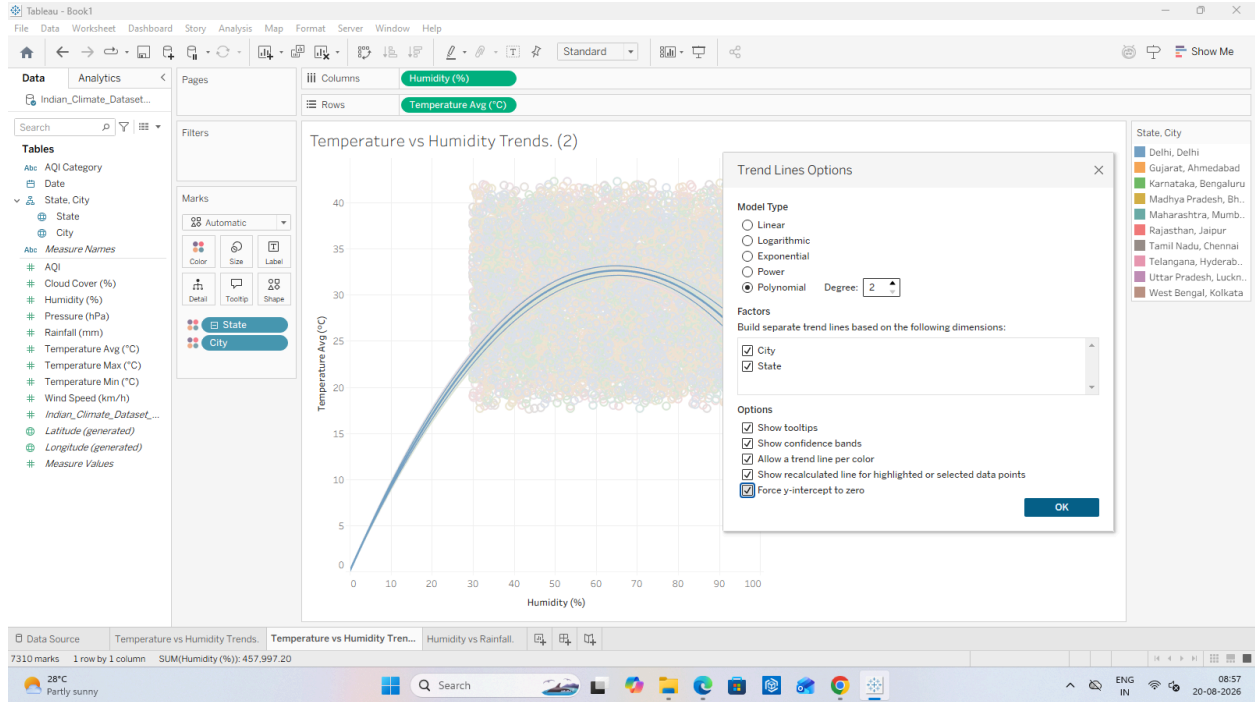
b) Customize and analyze trend models
OUTPUT:

Purvi.R.Karkera
T086

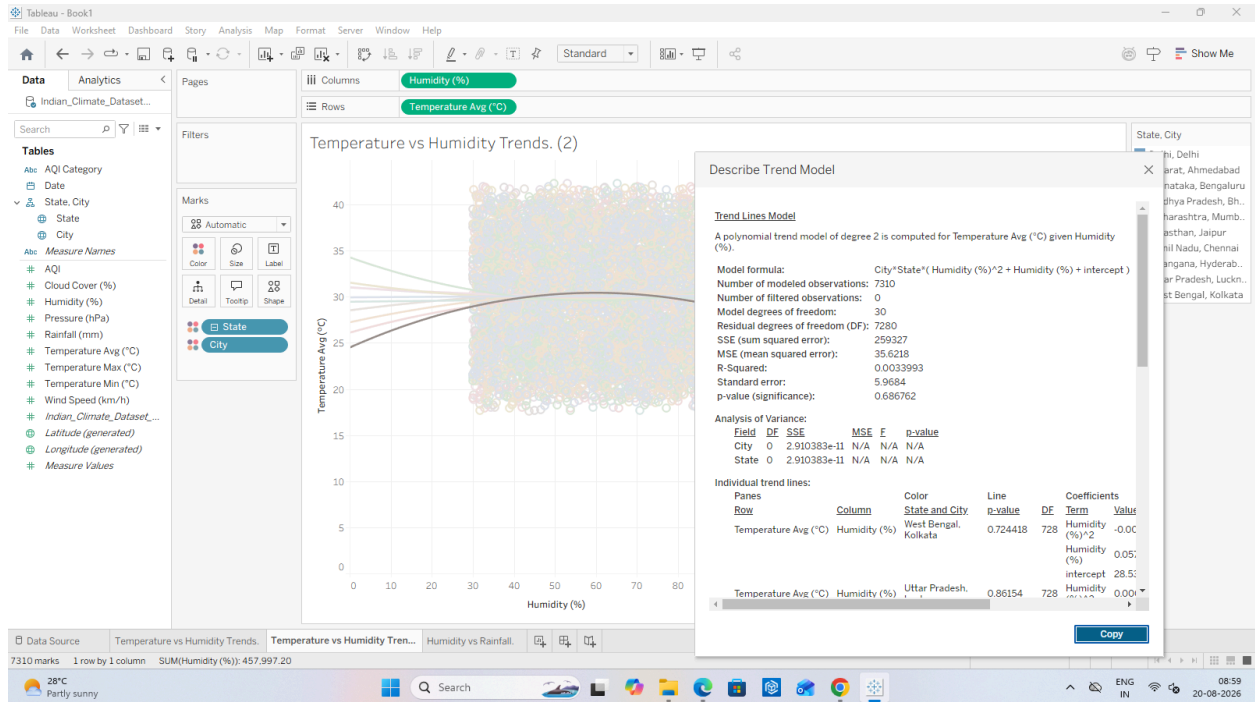
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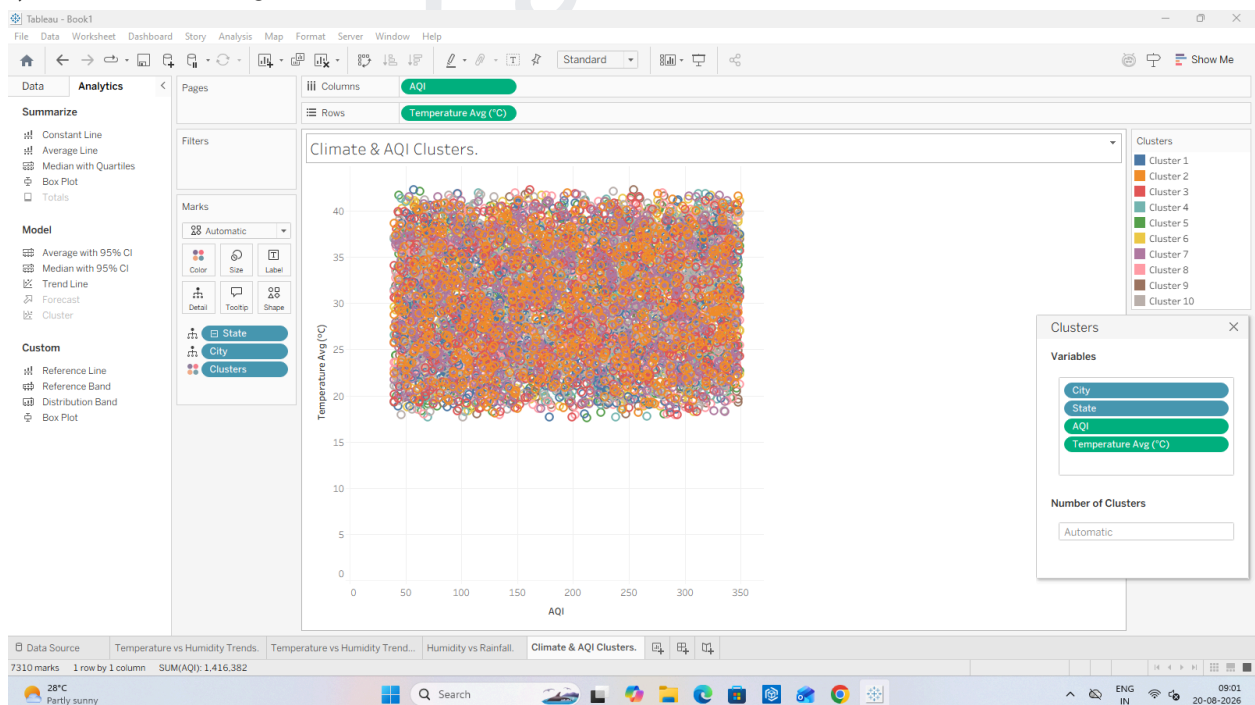
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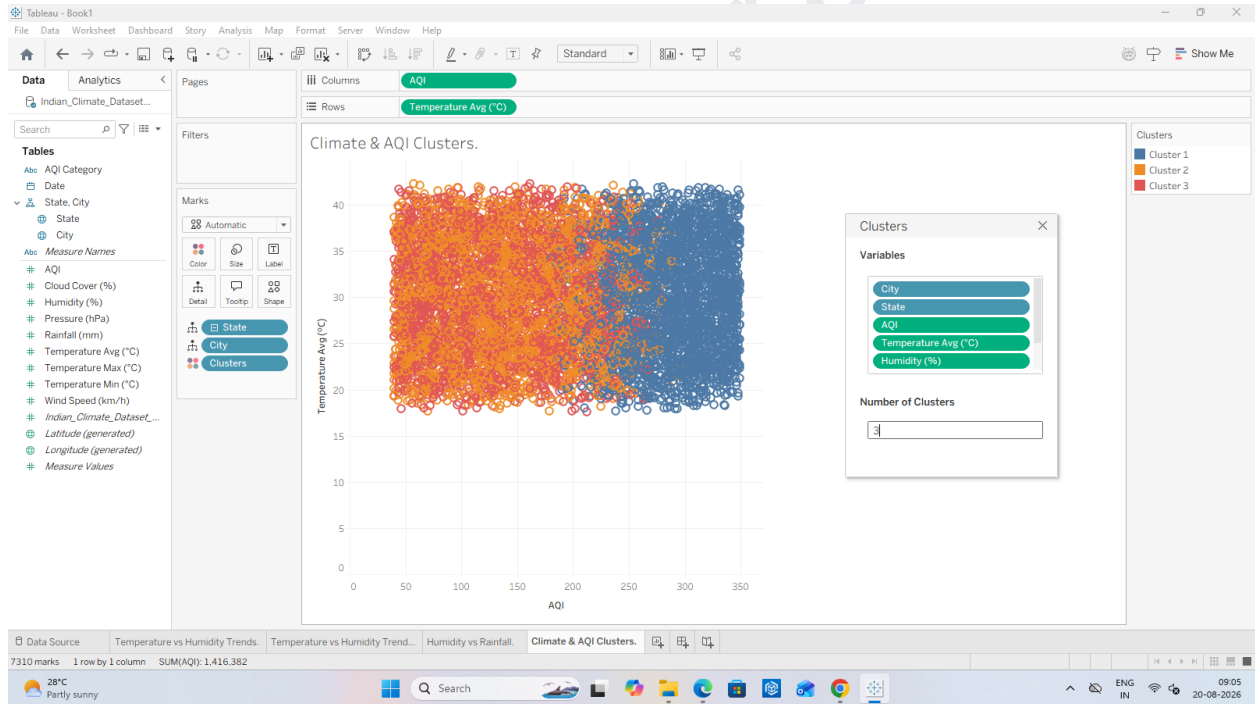
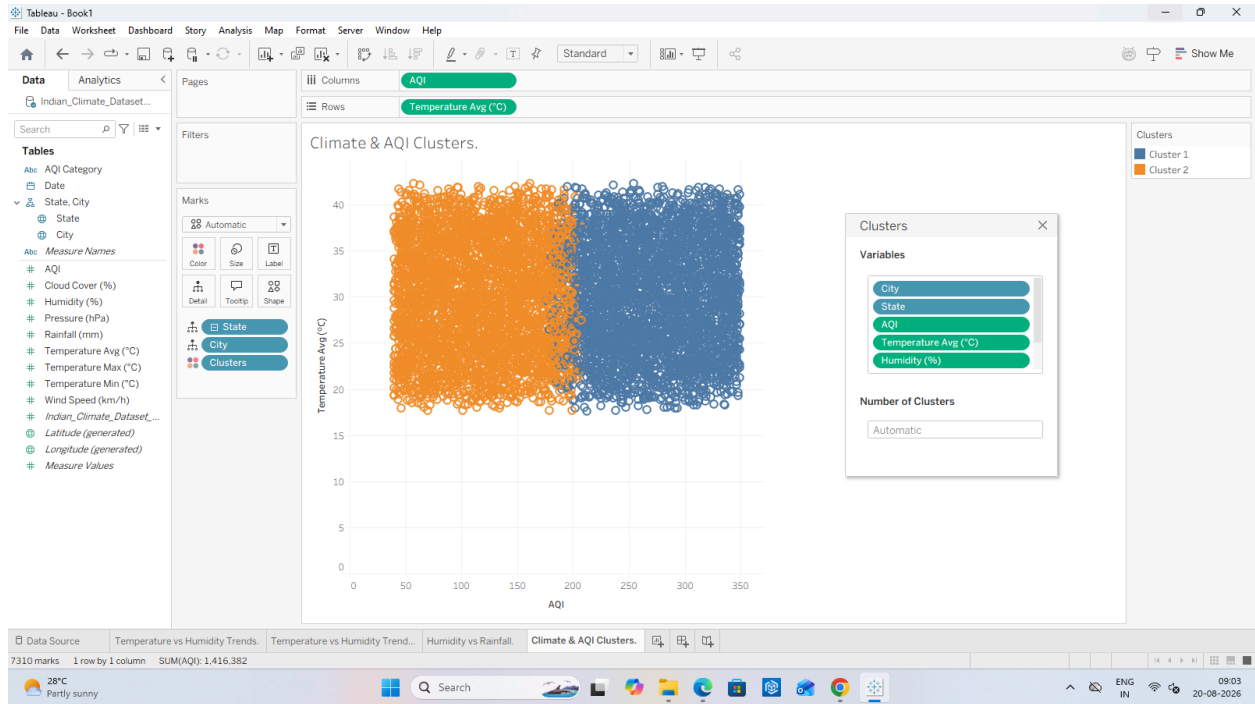
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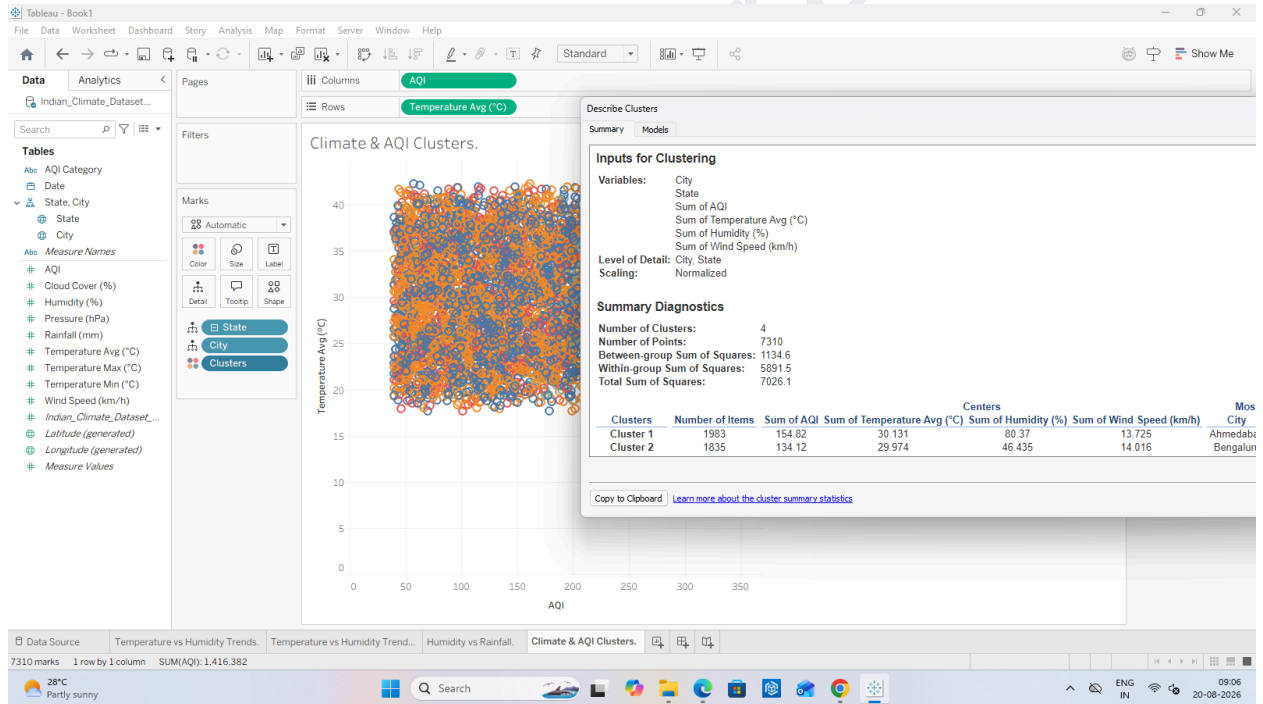
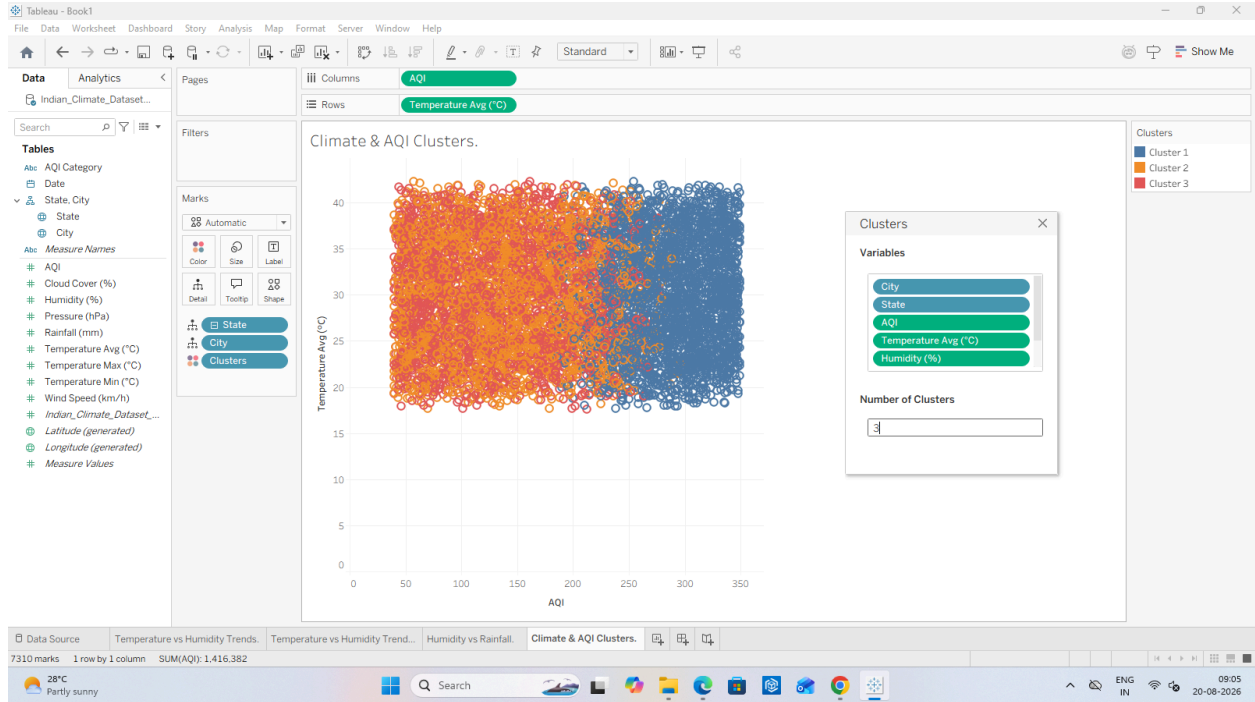
c) Perform clustering on datasets



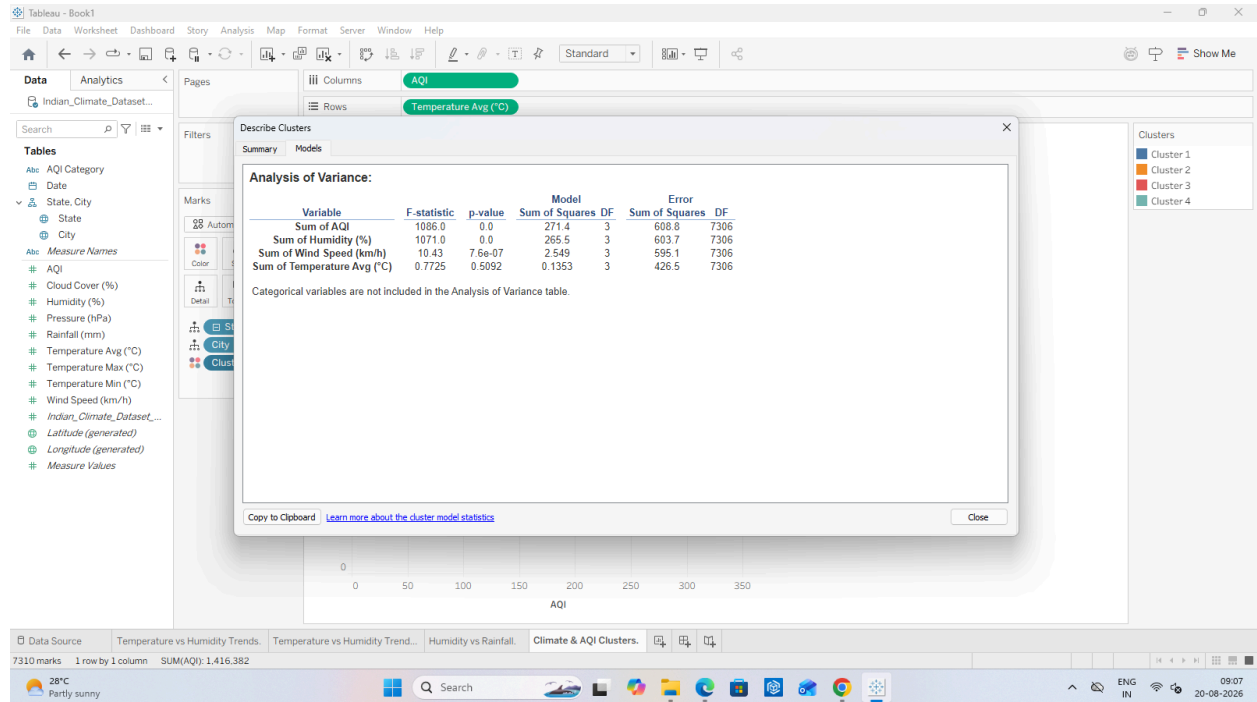
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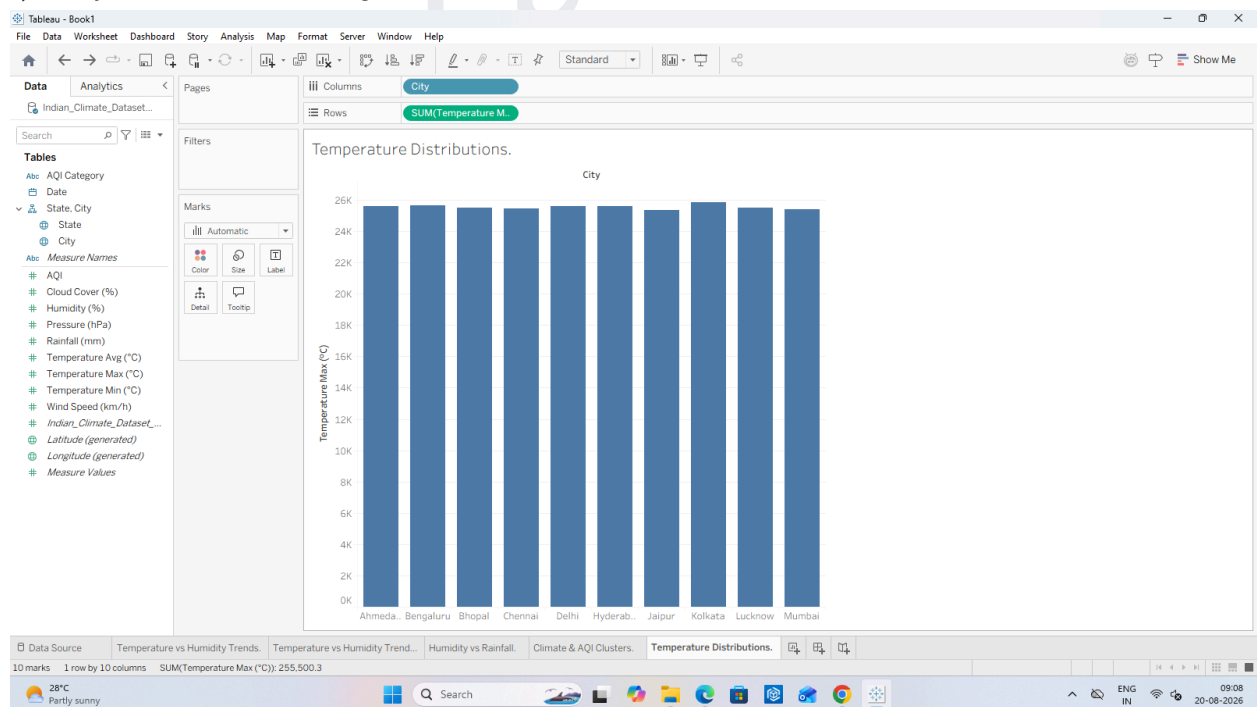
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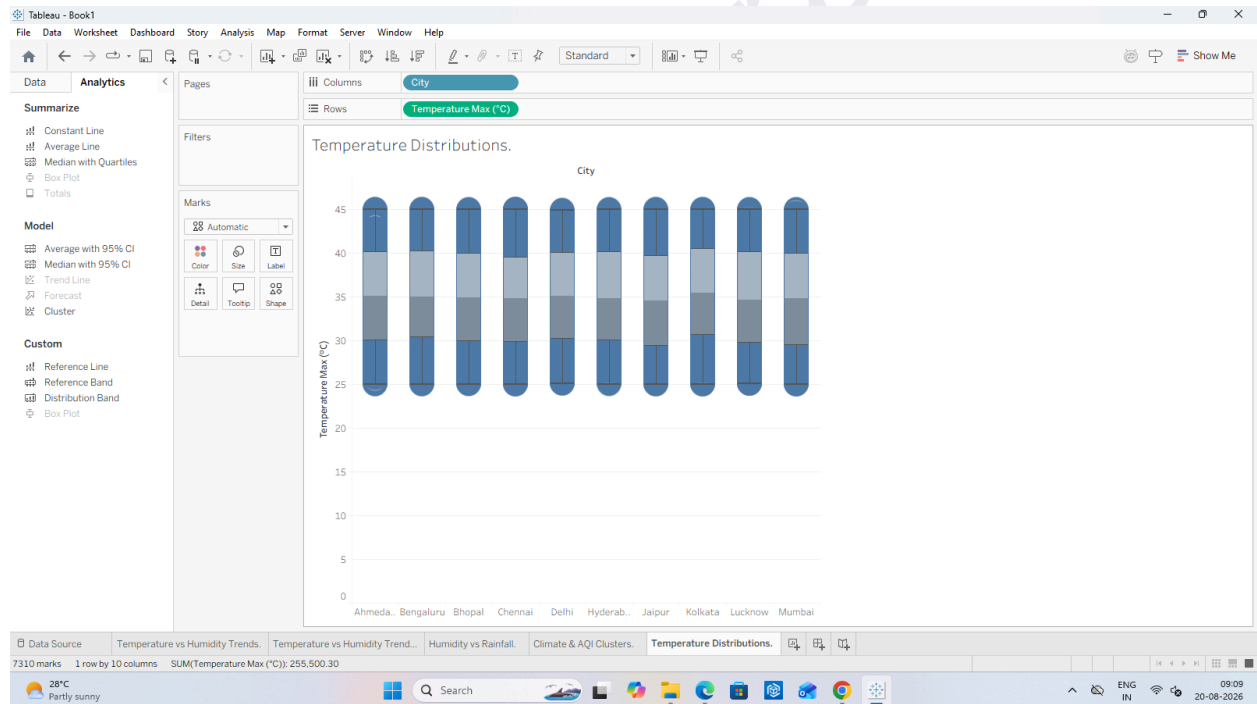
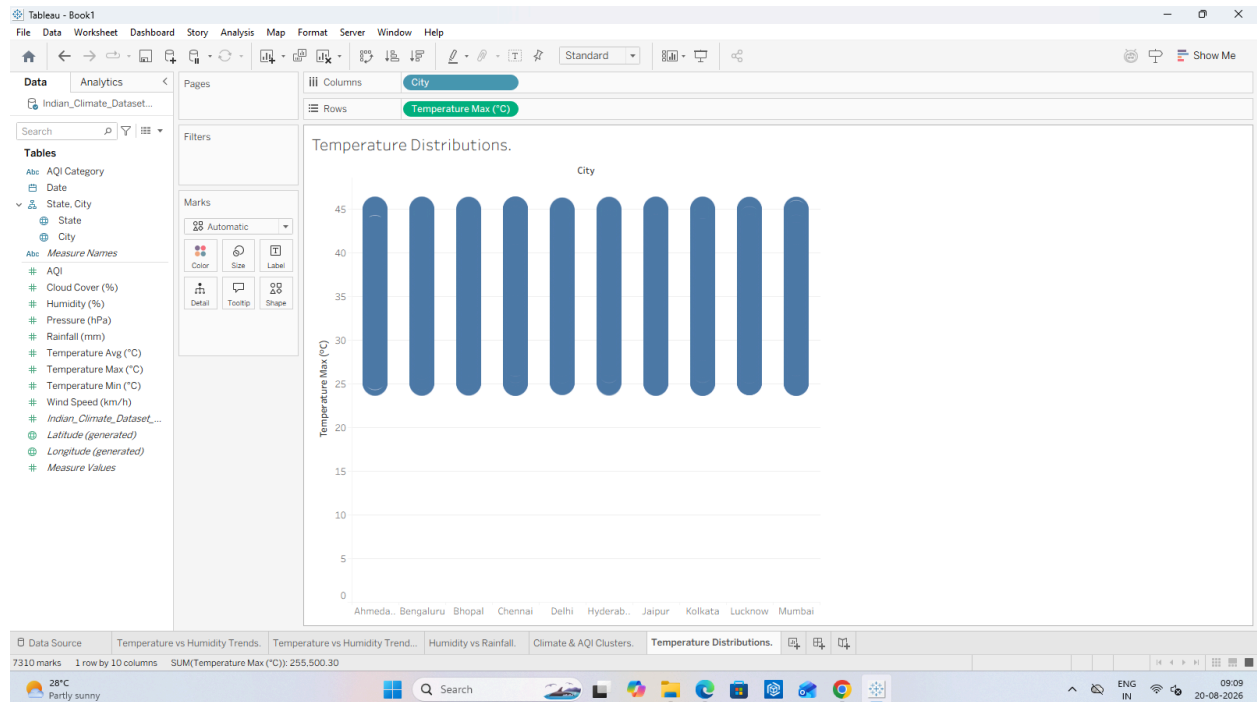
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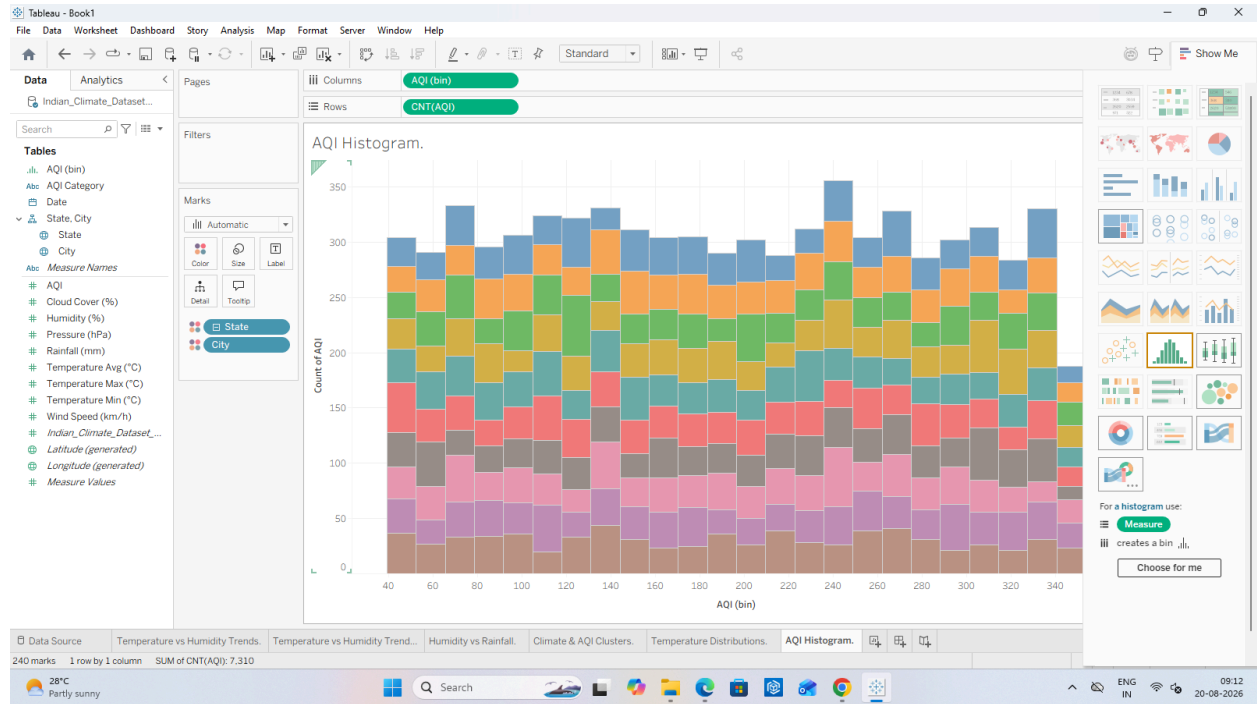
d) Analyze distributions using built-in tools



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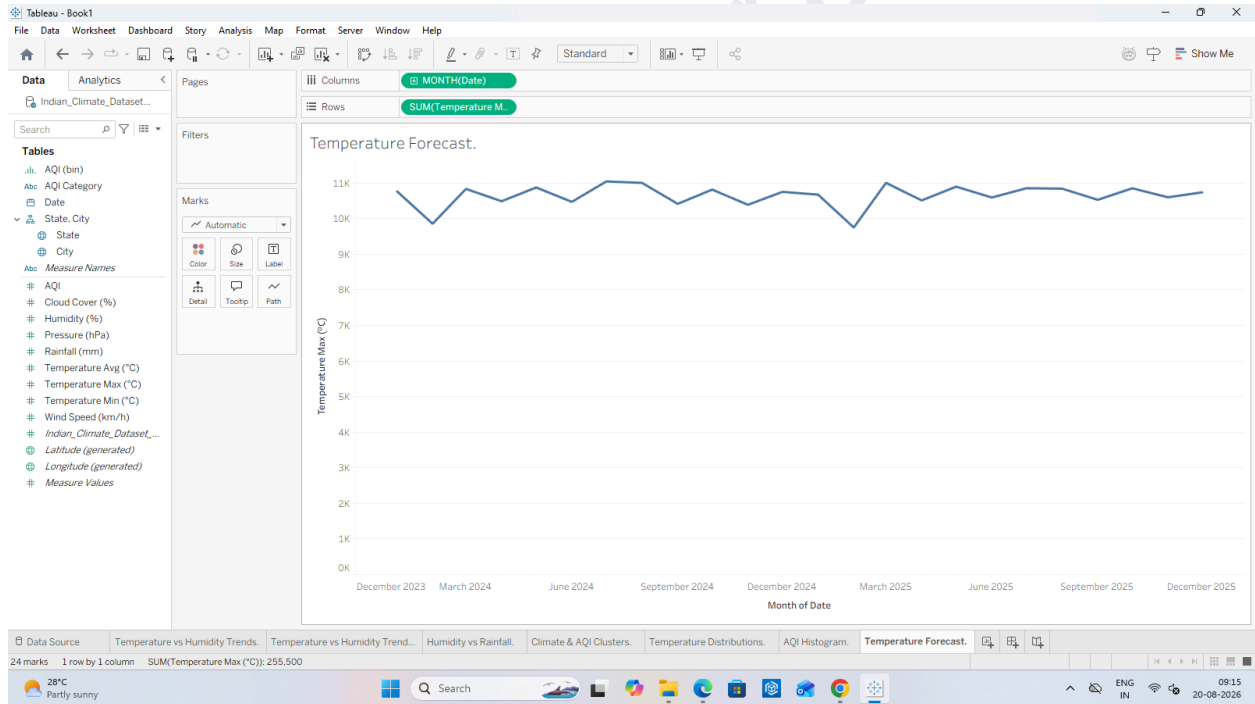
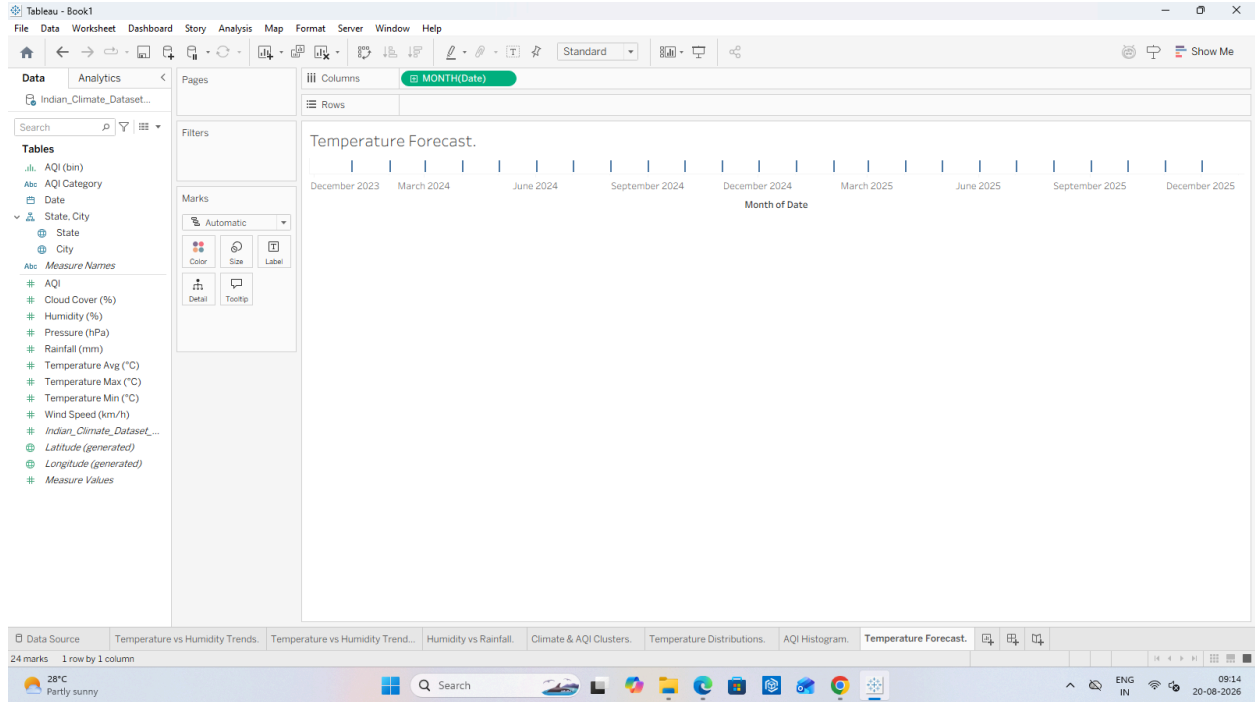


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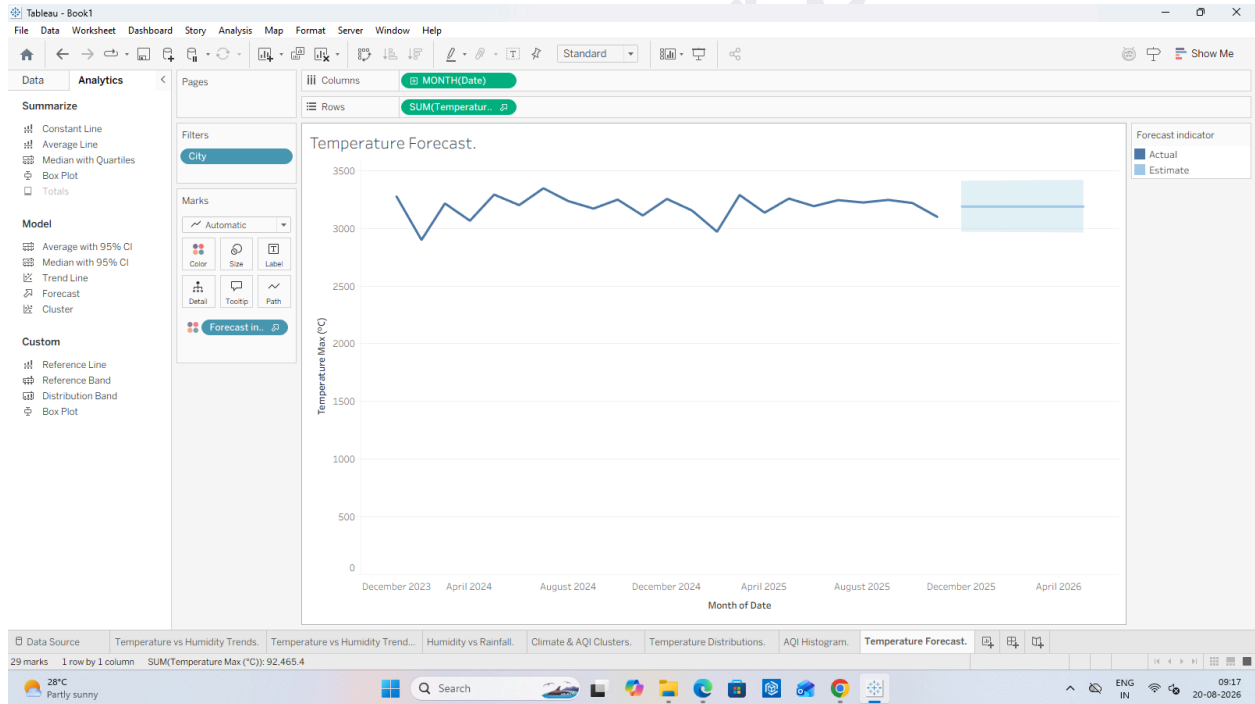
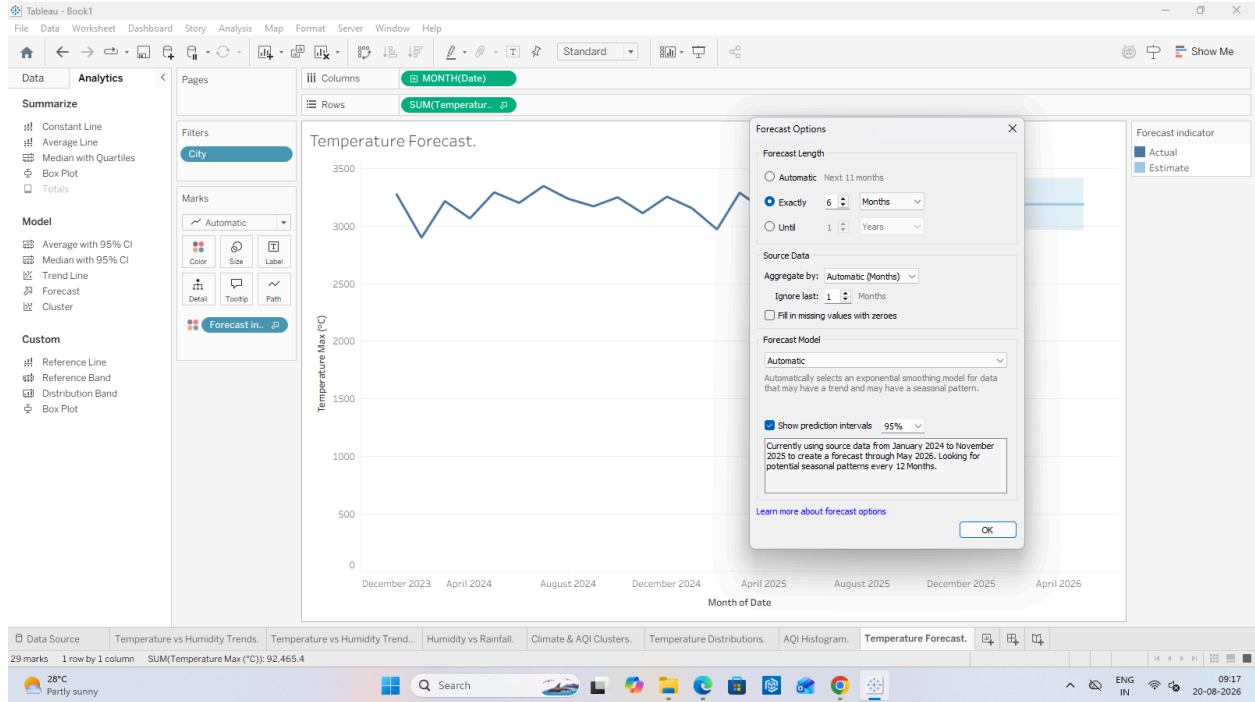


e) Apply forecasting techniques

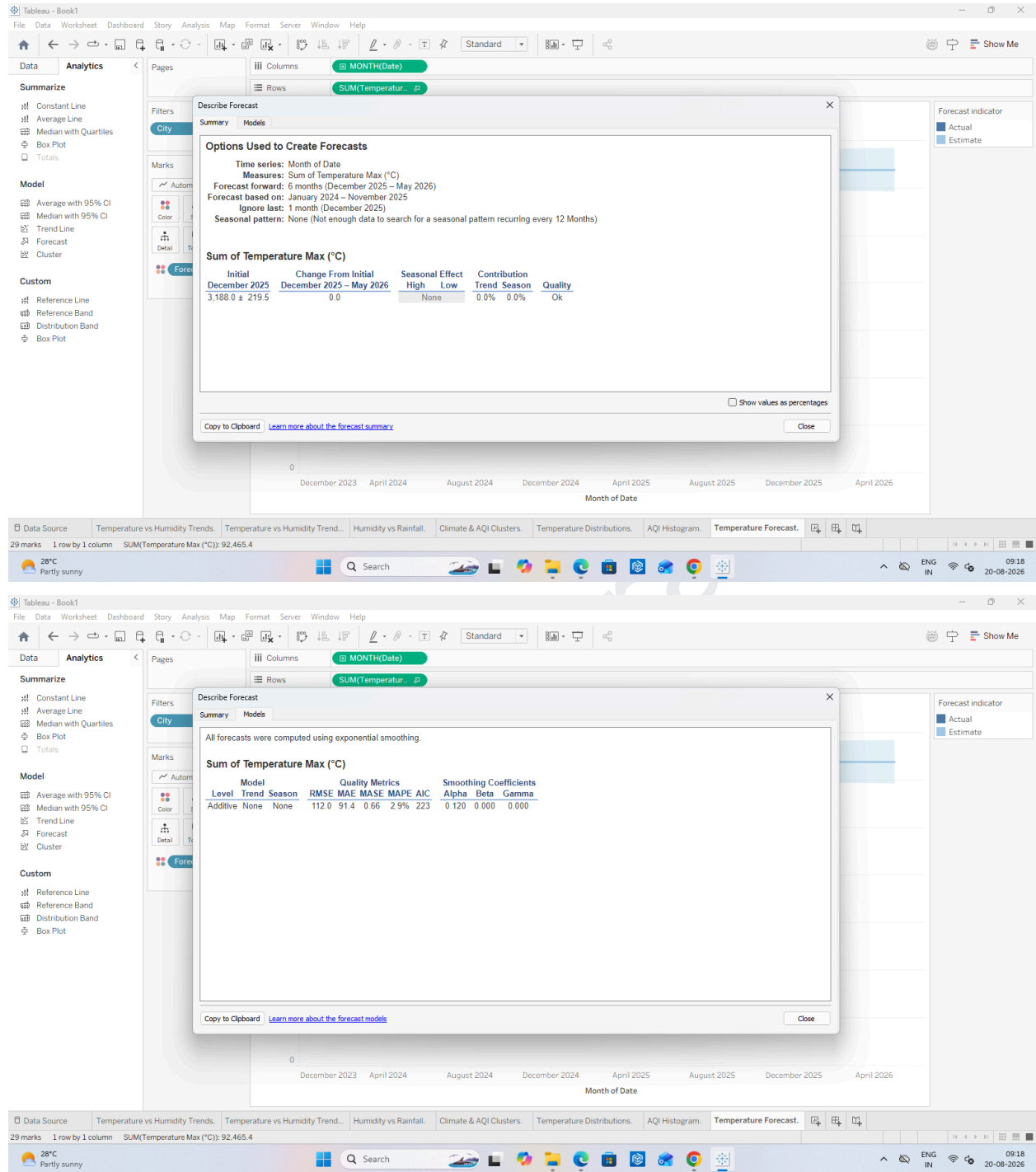
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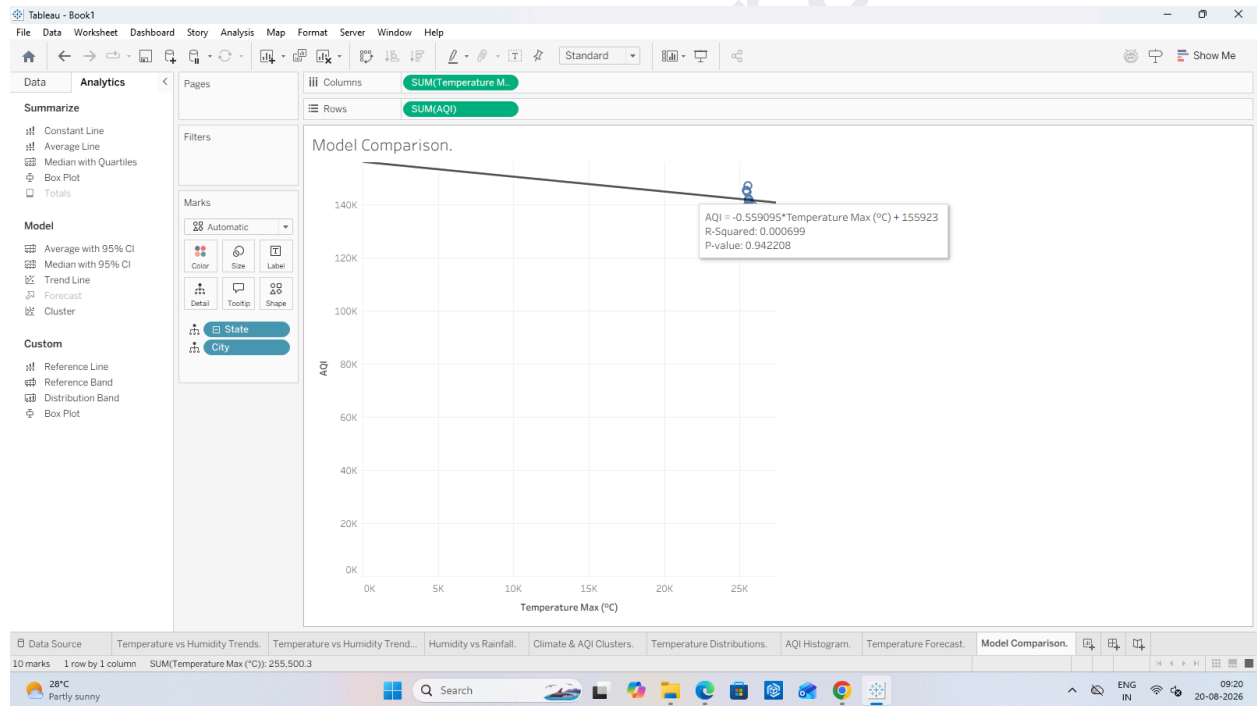
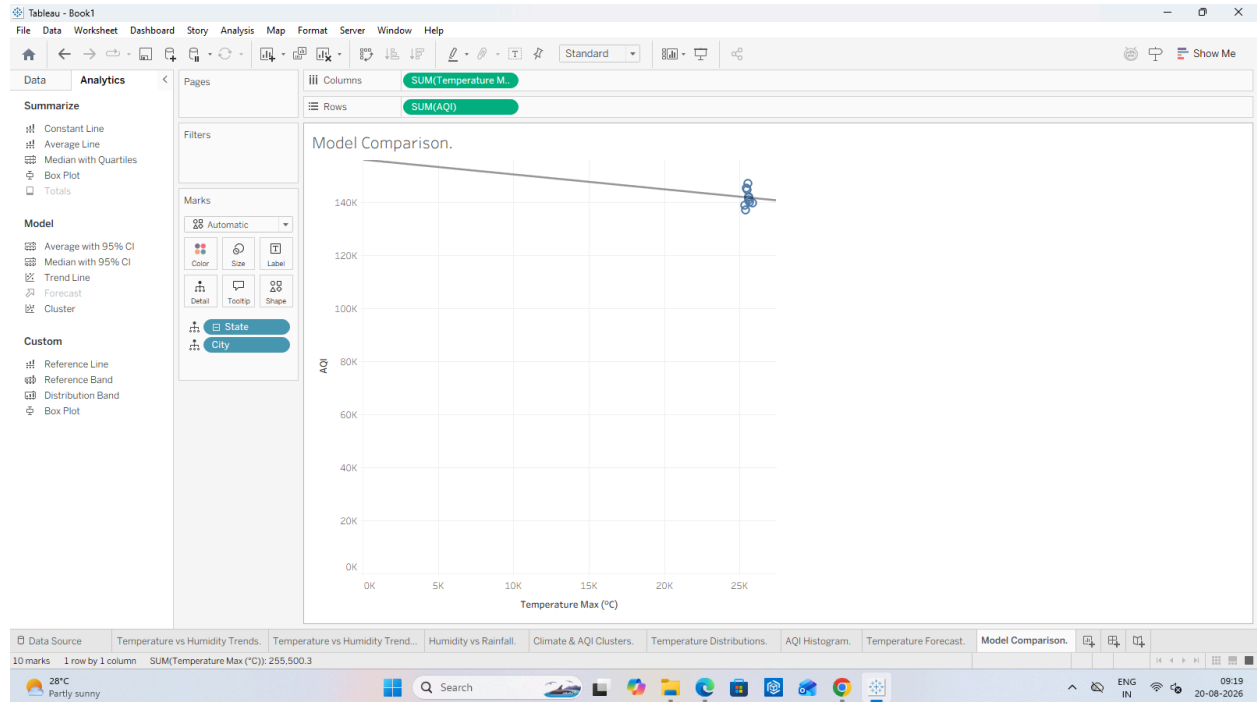


f) Compare different statistical models

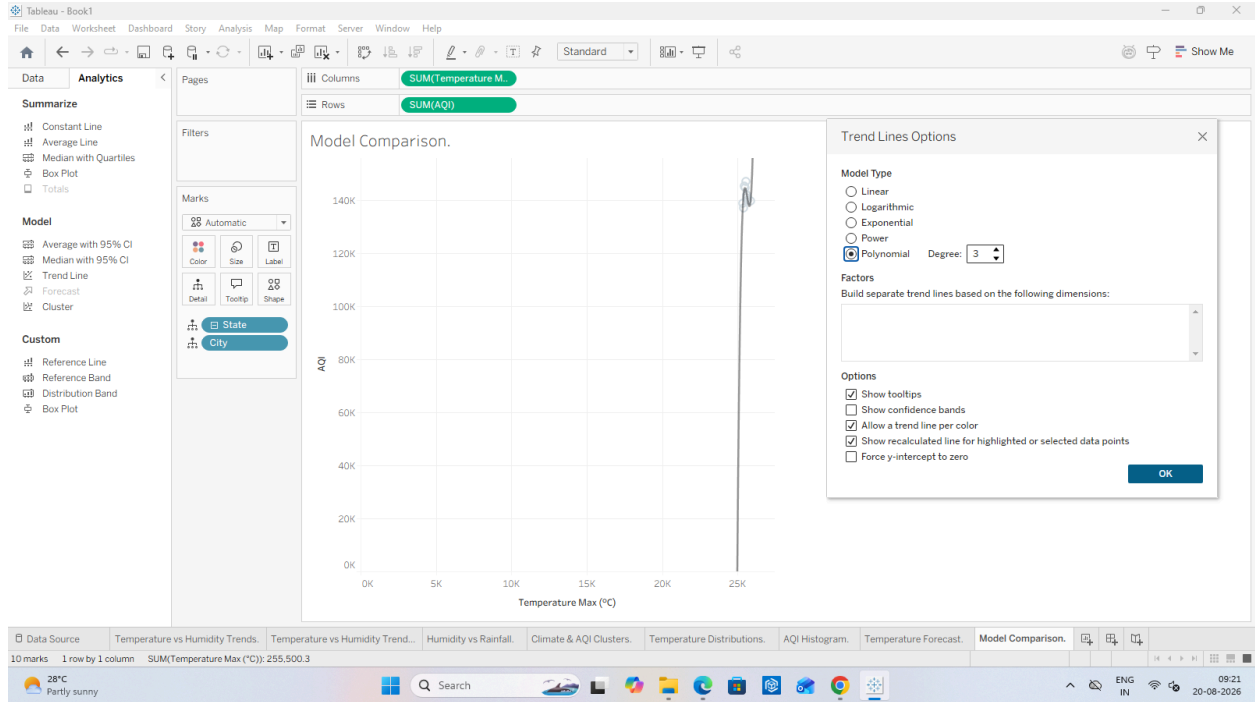
Purvi.R.Karkera

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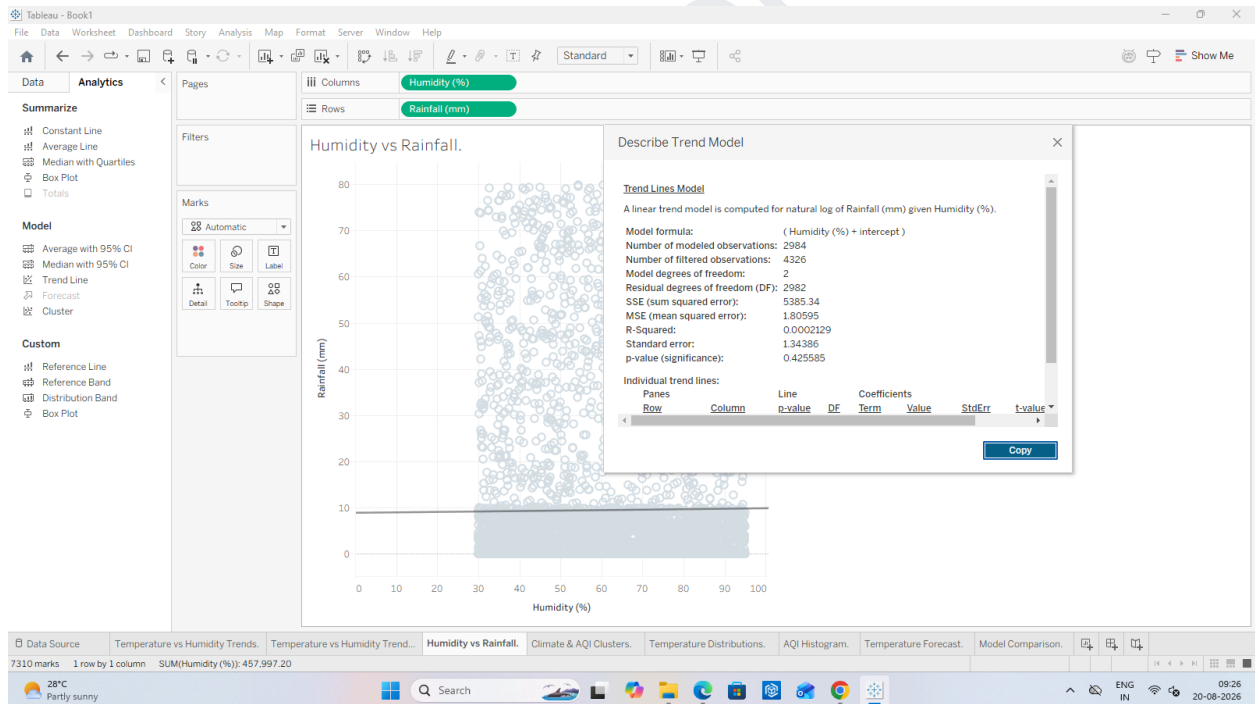
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g) Export and interpret statistical results



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```
Real estate.csv  diabetes_small.csv  INSTALL.txt  students.txt  Trend Lines Model  +
File Edit View H1 B I S G D View
Trend Lines Model
A linear trend model is computed for natural log of Rainfall (mm) given Humidity (%).
Model formula: ( Humidity (%) + intercept )
Number of modeled observations: 2984
Number of filtered observations: 4326
Model degrees of freedom: 2
Residual degrees of freedom (DF): 2982
SSE (sum squared error): 5385.34
MSE (mean squared error): 1.80595
R-Squared: 0.0002129
Standard error: 1.34386
p-value (significance): 0.425585
Individual trend lines:
Panels Line Coefficients
Row Column p-value DF Term Value StdErr t-value p-value
Rainfall (mm) Humidity (%) 0.425585 2982 Humidity (%) 0.0010491 0.0013166 0.796879 0.425585
Intercept 2.18398 0.0854115 25.5701 < 0.0001
```

Ln 20, Col 46 679 characters Plain text 100% Windows (CRLF) UTF-8

28°C Partly sunny Search 09:25 20-08-2026